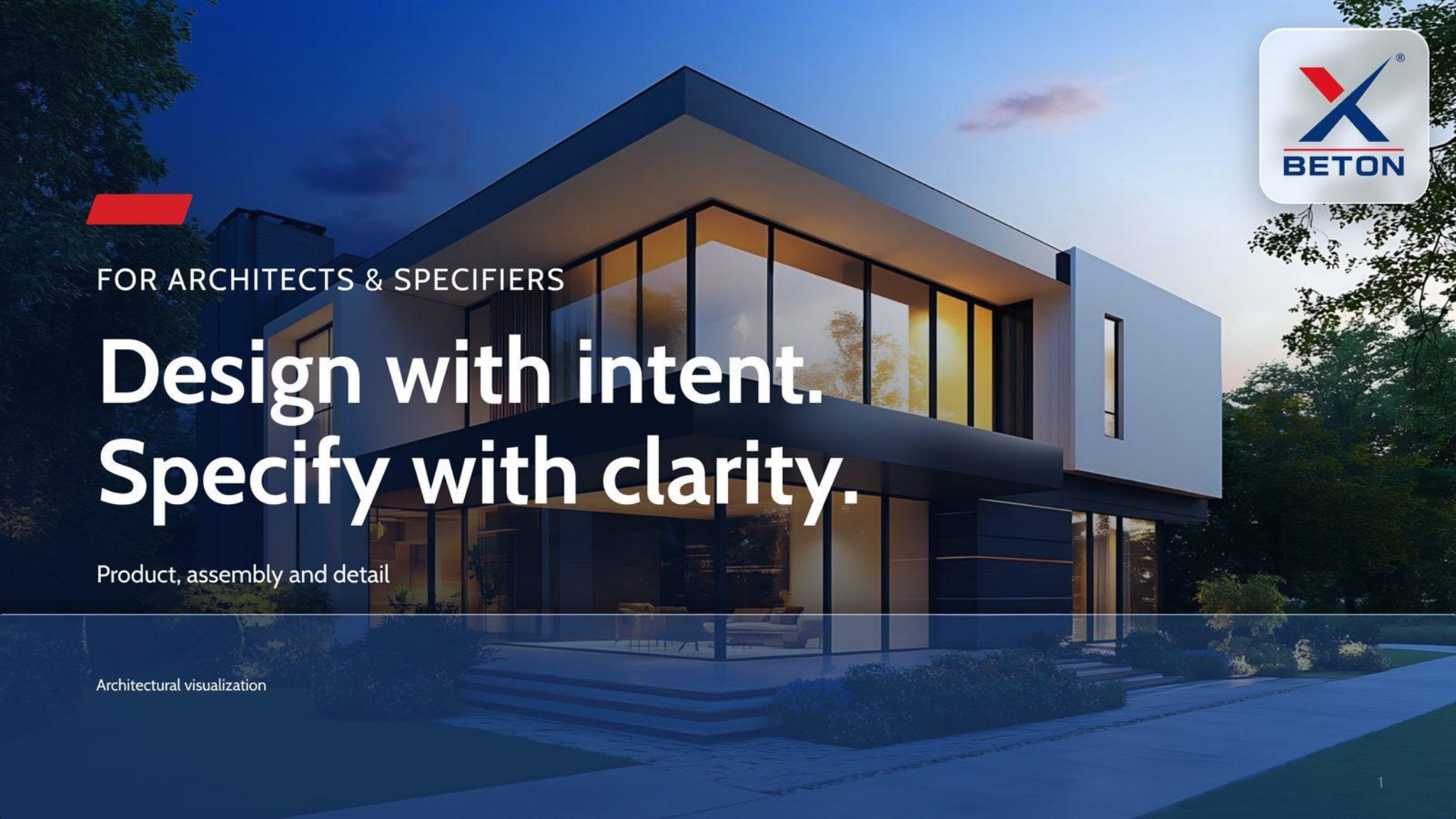
FOR ARCHITECTS & SPECIFIERS

Design with intent. Specify with clarity.

Product, assembly and detail



Architectural visualization

01 · APPLICATION

The specification starts with the application

Identify structural or nonstructural use. Define wall, floor, roof or cladding application, then coordinate fire, thermal, acoustic and moisture requirements.

Application	Requirement to establish	Evidence needed
Wall	Structural or nonstructural role, exposure	Product data and assembly reports
Floor or roof	Spans, loads, support conditions	Manufacturer design data, engineer review
Cladding	Finish, movement, moisture behavior	Finish system documentation
Fire, thermal, acoustic	Project-specific targets	Applicable test reports and calculations

Architectural inspiration. Scope the application before selecting a product.

Product families and design responsibility

Select the product with the responsible design professionals.



XBeton product image

AAC blocks

Masonry blocks and special shapes



XBeton product image

Reinforced panels

Reinforced panel products



Illustrative Bond & Shield application

Bond & Shield

Admixture and coating family

Design responsibility: the engineer of record and design professionals confirm the product for each application.

Material properties as an evaluation starting point

Supplier-reported values depend on product and density. Request the exact product data before design.

Property	FAQ range	Required project document
Dry density	25-40 lb/ft³	Product data sheet for the selected class
Compressive strength	290 to more than 1,090 psi	Test report with class, conditioning and current revision

Supplier-reported ranges, FAQ 2025 p. 6. Not design values. Product-dependent.

Architectural inspiration: Nicholas C. Metropolis Center background photo.



Fire performance and the evidence gap

6-hour fire rating [NEEDS VERIFY]

Owner-selected marketing claim. Assembly/test evidence pending.

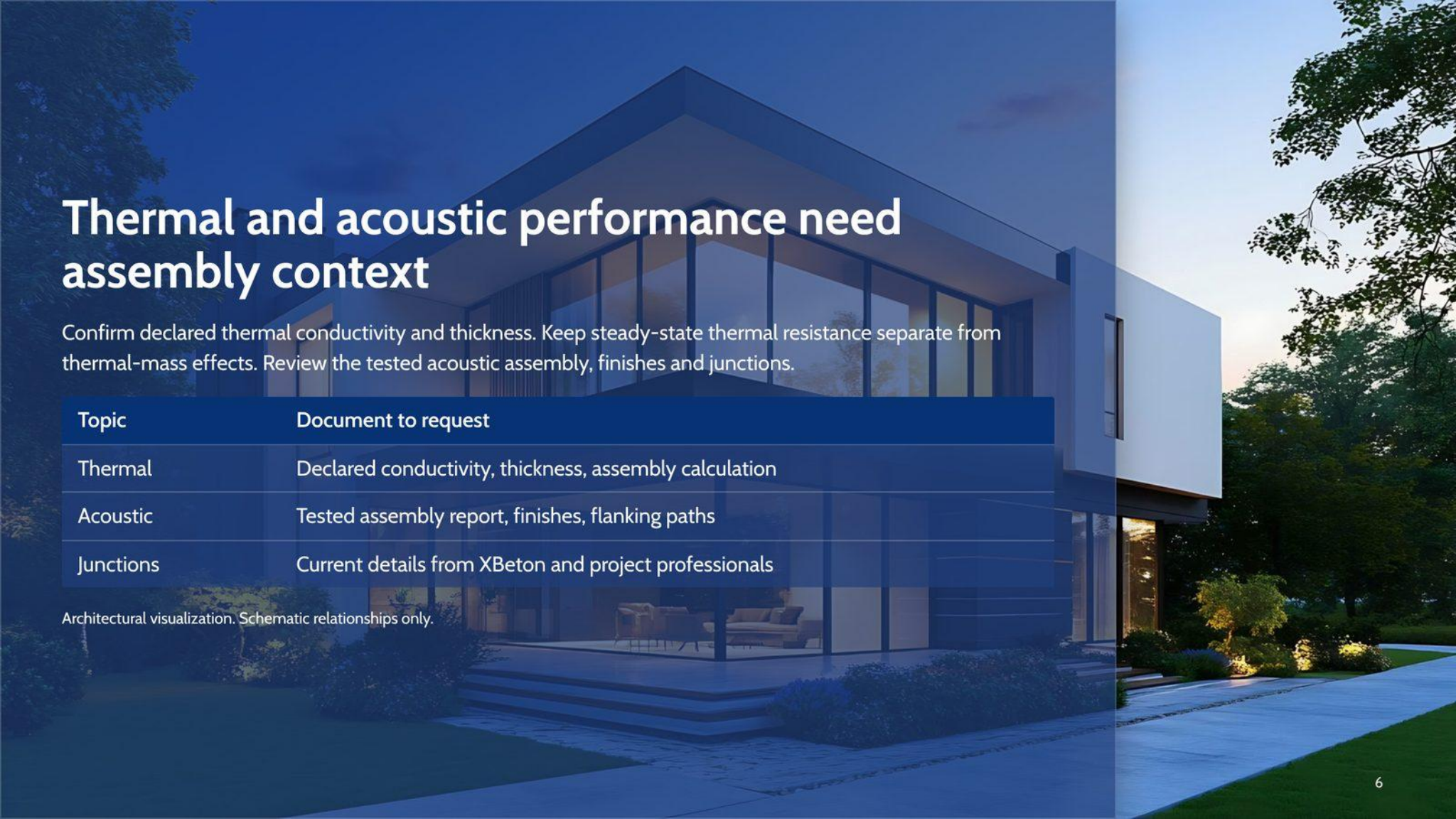
- [ ] Request the applicable assembly report and limitations.
- [ ] Confirm product, thickness, joints, openings and support conditions.

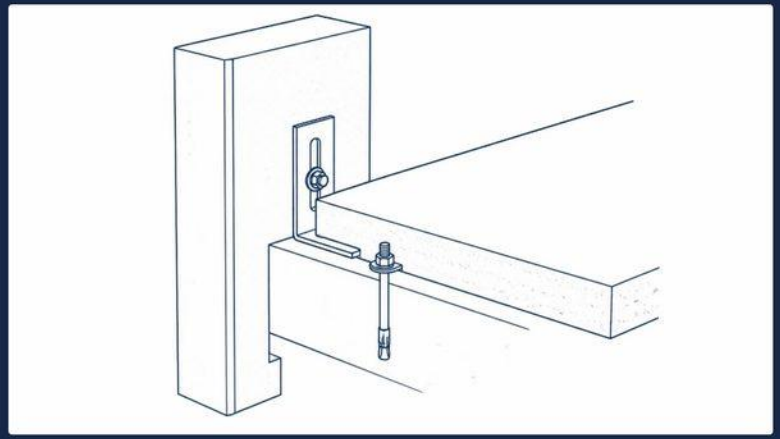
Thermal and acoustic performance need assembly context

Confirm declared thermal conductivity and thickness. Keep steady-state thermal resistance separate from thermal-mass effects. Review the tested acoustic assembly, finishes and junctions.

Topic	Document to request
Thermal	Declared conductivity, thickness, assembly calculation
Acoustic	Tested assembly report, finishes, flanking paths
Junctions	Current details from XBeton and project professionals

Architectural visualization. Schematic relationships only.





Illustrative panel connection schematic. Not a project detail.

Connections, openings and moisture detailing

Support

Structural load path and panel support

Openings

Door and window openings

Penetrations

MEP penetrations and attachments

Moisture

Vapor-compatible finish system

A clear submittal and compliance path

Compile current product identification and data, applicable test reports and evaluation documents, plus project drawings, specifications and shop drawings.

Document	Revision	Applicability
Product data	Confirm	Confirm
Assembly reports	Confirm	Confirm
Shop drawings	Confirm	Confirm
Environmental documentation	Confirm	Confirm

Review by the design team and authority having jurisdiction.

Applicability to be confirmed for each project.

Architectural inspiration: background photo, Nicholas C. Metropolis Center.



Sustainability documentation for the selected product

Request product-specific environmental information.

Evaluate operational and embodied impacts separately.

Confirm durability, finish and end-of-life assumptions.

AAC architecture reference. Image only; no environmental certification implied.

Request a technical review

Share application, design stage and location. Identify performance requirements and missing documents. Bring the exact assembly questions.

Product information

https://www.xbeton.com/

Project resources

https://build.xbeton.com/

Technical enquiry

https://www.xbeton.com/contactus

Specifier technical-review route: https://build.xbeton.com/specifiers#project

Architectural visualization.